KIRCHBERG ALGEBRAS AS FULL AMALGAMS OF STABLY FINITE ALGEBRAS

Abstract. We present a proposed proof that every separable nuclear unital simple purely infinite C*-algebra B admits a decomposition $B \cong A_0 *_D A_1$, where D, A_0, A_1 are separable nuclear simple unital stably finite C*-algebras and the common inclusions are unital and injective. The product carries the full universal norm. No UCT assumption is made on B. The construction combines finite-dimensional ordinary-graph amalgams with a nuclear residually finite-dimensional coefficient algebra KK-equivalent to B. Three positive embedding channels preserve the reduced coefficient KK-class, cancel its scalar summand, and enforce simplicity. A projection in the common algebra prescribes the unit, and unit-preserving KK classification supplies the final identification. If correct, the argument answers the question in Suzuki's Remark 1.5 for all unital Kirchberg algebras.

1. Statement and conventions

Theorem 1 (Proposed main theorem)**.** *Every unital Kirchberg algebra B has a unital isomorphism*

$$B \cong A_0 *_D A_1$$

where D, A_0, A_1 are separable nuclear simple unital stably finite C-algebras and both inclusions $D \to A_j$ are unital and injective. The product is the full unital amalgamated free product.*

A Kirchberg algebra means separable, nuclear, simple and purely infinite; no UCT assumption is implicit. All tensor products are minimal except where the maximal universal property is explicitly used. Matrix multiplicity matrices act on column vectors, with target blocks as rows. A minimal matrix projection is the K_0 generator. $SX = C_0((0,1)) \otimes X$. We write Kasparov composition from left to right: $x \otimes_Y y$ first applies $x \in \mathrm{KK}(X, Y)$ and then $y \in \mathrm{KK}(Y, Z)$.

The question is posed in [1, Remark 1.5, p. 2]; the constituent requirements are those of its Corollary 1.4. Nuclearity of the three constituents is an additional conclusion here. No expectation onto the limit common algebra is required or asserted. The proof uses UCT only for explicitly identified finite graph algebras, never for B or the coefficient algebra.

2. An ordinary graph as a full finite-dimensional amalgam

Let A be an $r \times r$ nonnegative integer matrix with no zero column, and let $k, \ell \in \mathbb{Z}^r$ have strictly positive entries. Define

$$\begin{aligned} D(A; k, \ell) &= \bigoplus_{v=1}^{r} M_{k_v} \ \oplus \ \bigoplus_{v=1}^{r} M_{\ell_v}, \\ F_0(A; k, \ell) &= \bigoplus_{v} M_{k_v + \ell_v}, \\ F_1(A; k, \ell) &= \bigoplus_{v} M_{k_v + \sum_w A_{vw} \ell_w}. \end{aligned}$$

The common inclusions have multiplicity matrices $[I\ I]$ and $[I\ A]$. Write $P(A; k, \ell) = F_0 *_D F_1$.

Date: Version 2, 7 October 2026.

Let E_A have A_{vw} edges from w to v; its usual source-row adjacency matrix is A^t. Write its vertex projections p_v and edge partial isometries s_{wvt}, where $1 \le t \le A_{vw}$. Thus

$$s_{wvt}^* s_{wvt} = p_v, \qquad \sum_{v,t} s_{wvt} s_{wvt}^* = p_w.$$

Distinct edges have orthogonal range projections. With $m = k + \ell$ and $M = \sum_v m_v$, put $q = \mathrm{diag}(p_v \text{ repeated } m_v \text{ times})$ in $M_M(C^*(E_A))$.

Lemma 2. *There is a unital isomorphism*

$$P(A; k, \ell) \cong q M_M(C^*(E_A)) q.$$

The projection q is full. The minimal projection of the vth F_0 summand corresponds to $[p_v]$ in K_0, and the product unit corresponds to the vector $k + \ell$.

Proof. Index the vth F_0 matrix units E^v by $P_{v,a}$ and $Q_{v,b}$, with $a \le k_v$ and $b \le \ell_v$. Index the vth F_1 units F^v by $P_{v,a}$ and $Q_{w,t,b}$, with $t \le A_{vw}$, $b \le \ell_w$. The common relations are

$$E^v_{P_{v,a},P_{v,b}} = F^v_{P_{v,a},P_{v,b}}, \qquad E^w_{Q_{w,b},Q_{w,c}} = \sum_{v,t} F^v_{Q_{w,t,b},Q_{w,t,c}}.$$

In the proposed corner send E^v_{ij} to $e_{ij} \otimes p_v$. Writing $a_v = P_{v,1}$, define

$$V^v_{P_{v,a}} = e_{P_{v,a},a_v} \otimes p_v, \qquad V^v_{Q_{w,t,b}} = e_{Q_{w,b},a_v} \otimes s_{wvt}.$$

Within each v these have common initial projection $e_{a_v,a_v} \otimes p_v$ and orthogonal final projections. Final projections for different v are also orthogonal. Their total sum is q by the graph relations. Hence $V^v_\alpha (V^v_\beta)^*$ defines a unital representation of F_1, agreeing with that of F_0 on D. The full universal property gives a homomorphism Ψ to the corner.

Conversely put $p^P_v = E^v_{a_v,a_v}$, $p = \sum_v p^P_v$, and

$$t_{wvt} = E^w_{P_{w,1},Q_{w,1}} F^v_{Q_{w,t,1},P_{v,1}} \in pPp.$$

Matrix multiplication gives the graph relations, with initial projection p^P_v, orthogonal ranges for distinct edges, and range sum p^P_w over edges starting at w. The graph universal property [2, Theorem 1.2] gives $\theta : C^*(E_A) \to pPp$. For a matrix entry $x_{ij} \in p_v C^*(E_A) p_w$ set

$$\Theta([x_{ij}]) = \sum_{i,j} E^v_{i,a_v} \theta(x_{ij}) E^w_{a_w,j}.$$

The orthogonal matrix units make this a *-homomorphism from the corner. Both composites fix the E units. They also fix the graph edges in the chosen vertex corner and the mixed units

$$F^v_{Q_{w,t,b},P_{v,a}} = E^w_{Q_{w,b},P_{w,1}} t_{wvt} E^v_{P_{v,1},P_{v,a}}.$$

These mixed units, their adjoints and products recover all F units; the graph edges and the E units generate the inflated graph corner. Thus both composites are identities. Each vertex occurs in q, so q is full. The generator and unit statements follow from these explicit maps. □

We henceforth set $A = I + B$, where every entry of B is a positive integer. The finite graph is strongly connected, has no sinks, and has at least two loops at every vertex. It is cofinal and every cycle has an exit. Its algebra is simple and purely infinite by [2, Corollary 3.11], and nuclear by the ordinary graph nuclearity statement in [3, Remark 3.10]. The full corner in Lemma 2 is therefore a unital Kirchberg algebra. No separated-graph nonnuclear full product is substituted for this ordinary graph algebra.

Finite-dimensional inclusions have unital conditional expectations. The sequence from [4, Theorem 4.1] gives the integer map

$$\phi(x, y) = (x + y, -x - Ay), \qquad K_0(P) = \mathrm{coker}\, B, \quad K_1(P) = \ker B, \quad [1_P] = [k + \ell].$$

We identify $y \in \ker B$ with $(-y, y) \in \ker \phi$. The K_0 generators are the minimal projections of F_0. These conventions will fix all bonding maps, rather than merely the abstract groups.

3. Lifting arbitrary graded maps by positive diagrams

Let source and target matrices be $B = A - I$ and $B' = A' - I$, of sizes r and r'. Suppose integer matrices S, H satisfy $SB = B'H$. For any $r' \times r$ integer matrix Z, define

$$T = \begin{pmatrix} S - Z & S - H - ZA \\ Z & H + ZA \end{pmatrix}, \qquad R = S + B'Z. \tag{1}$$

Direct multiplication gives

$$[I\ I]T = S[I\ I], \qquad [I\ A']T = R[I\ A]. \tag{2}$$

When T, S, R are strictly positive, set $\binom{k'}{\ell'} = T\binom{k}{\ell}$. These identities give actual unital finite-dimensional embeddings of multiplicities T, S, R. Fix the standard common-algebra embedding. The two representations of D in each target factor have equal irreducible multiplicities; match equal copies with a permutation unitary and conjugate the whole factor embedding. The squares then commute literally. The induced full-product homomorphism is injective because its source is simple and it is unital.

The maps on K-theory are $[S]$ on the cokernel and H on the kernel. Indeed the latter follows directly from $T(-y, y) = (-Hy, Hy)$ when $By = 0$, and the former from the first-factor minimal projections. This also follows from naturality of the cone sequence.

Lemma 3 (Simultaneous positive lifting)**.** *Let B, B'_0 be strictly positive square integer matrices, with target size at least two. For any finite list of prescribed pairs*

$$f_{0,c} : \operatorname{coker} B \to \operatorname{coker} B'_0, \qquad f_{1,c} : \ker B \to \ker B'_0,$$

one can replace the target presentation by $B' = UB'_0$, with U nonnegative unimodular, and find strictly positive matrices T_c, S_c, R_c as in (1), realizing each pair under the induced target identifications. All channels use the same B'.

Proof. First lift $f_{0,c}$ to an integer matrix $S_{0,c}$ on the free presentations. The columns of $S_{0,c}B$ lie in $\operatorname{im} B'_0$, so choose $H_{0,c}$ with $S_{0,c}B = B'_0 H_{0,c}$. The difference $f_{1,c} - H_{0,c}|_{\ker B}$ lands in $\ker B'_0$. Since $\mathbb{Z}^r / \ker B \cong \operatorname{im} B$ is free, $\ker B$ is a direct summand of $\mathbb{Z}^r$. Extend the difference to a map into $\ker B'_0$ and add it to $H_{0,c}$. This realizes both prescribed maps.

Let J denote an all-ones matrix of the appropriate rectangular size. For sufficiently large positive integers t_c, the matrices

$$\overline{S}_c = S_{0,c} + B'_0(t_c J), \qquad \overline{H}_c = H_{0,c} + (t_c J)B$$

are strictly positive. These additions preserve the chain equation and both induced group maps. Set $Z_c = J$ and $W_c = \overline{H}_c + JA$. Choose a common nonnegative unimodular U whose every row sum exceeds every entry of every W_c and exceeds one. Such matrices exist in size $r' \geq 2$: add a large multiple of row two to row one of the identity, and then a large multiple of the new row one to every other row. Each operation has determinant one.

Put $B' = UB'_0$, $S_c = U\overline{S}_c$, and $H_c = \overline{H}_c$. Every entry of S_c is at least the corresponding row sum of U. Thus $S_c - J$ and $S_c - W_c$ are strictly positive. The other blocks J, W_c and $R_c = S_c + B'J$ are strictly positive as well. Left multiplication by U changes neither the kernel nor the cokernel up to the stated identification and does not alter either prescribed map. □

For multiple channels, the dimensions are defined with $T_{\mathrm{tot}} = \sum_c T_c$, not with any single T_c. This gives a useful nonunital version of the construction. In each target common summand put the different channel blocks in disjoint coordinate subspaces. Their units are orthogonal

projections $q_c \in D'$ with $\sum_c q_c = 1$. The channel factor maps have units $i'_j(q_c)$. Their restrictions to D match the chosen channel embedding, by the same matrix representation equivalence as above. Hence the universal maps are

$$\gamma_c : P \longrightarrow q_c P' q_c.$$

They are nonzero and injective, and their ranges are orthogonal. Each q_c is full in D' because every channel multiplicity is positive. Their sum is a unital product map. The cone maps and boundary maps are natural also for these nonunital homomorphisms, so the individual channel maps still induce S_c, H_c as above. No compatibility of conditional expectations is required for this naturality: the cone map is the pointwise map on its defining continuous functions.

4. A nuclear residually finite-dimensional KK model

An algebra is residually finite-dimensional (RFD) if finite-dimensional *-representations separate its points. A separable RFD algebra has a countable separating family: select representations detecting a countable dense subset, with arbitrarily good norm bounds, and take their union. RFD passes to subalgebras and to minimal unitization. No UCT hypothesis is assumed or used for the nuclear RFD coefficient algebra.

Lemma 4. *For every separable nuclear algebra B, there is a separable nuclear RFD algebra F and an invertible $\xi \in \mathrm{KK}(F, B)$.*

Proof. The construction in [7, Lemma 2.4 and proof of Theorem 1.2] gives a separable nuclear RFD extension

$$0 \longrightarrow I = \bigoplus_n M_{r(n)} \xrightarrow{\theta} H \longrightarrow SB \longrightarrow 0.$$

For clarity, apply its quasidiagonal pullback construction to $\widetilde{SB}$, obtaining an algebra inside $\prod_n M_{r(n)}$, and take the inverse image of SB. The quotient is nuclear, so the extension is semisplit; the ideal and quotient are nuclear, so H is nuclear. It is RFD as a subalgebra of the matrix product.

Let

$$F = C_\theta = \{(f, x) \in C_0([0,1), H) \oplus I : f(0) = \theta(x)\}.$$

The extension $0 \to SH \to F \to I \to 0$ proves nuclearity. The two ambient summands are RFD, so F is RFD. It is separable. The mapping-cone triangle of the semisplit extension above identifies C_θ up to KK-equivalence with $S(SB)$. Bott periodicity supplies ξ. The hypothesis $K_*(B) = 0$ used later in Dadarlat's proof is not used in this construction or mapping-cone equivalence. □

Fix F and ξ for the target B. Put $E = \widetilde{F}$, with inclusion $\iota : F \to E$, scalar quotient $\varepsilon : E \to \mathbb{C}$, and scalar section $\eta : \mathbb{C} \to E$. Use the external unitization if necessary, so that $0 \to F \to E \to \mathbb{C} \to 0$ is always split. Split exactness gives a class $r \in \mathrm{KK}(E, F)$ such that

$$[\iota] \otimes_E r = 1_F, \qquad r \otimes_F [\iota] = e := 1_E - [\varepsilon] \otimes_{\mathbb{C}} [\eta]. \tag{3}$$

In particular $e^2 = e$. Choose unital finite-dimensional representations

$$\sigma_h : E \longrightarrow M_{q_h}(\mathbb{C})$$

whose every tail separates points. For example, repeat each member of a countable separating family infinitely often. Their direct sum is faithful, so $\{\mathrm{id}_X \otimes \sigma_h\}$ separates $X \otimes E$ for every X: represent X faithfully and use the block decomposition of the faithful spatial representation. No homotopy condition on σ_h is needed.

5. The three actual coefficient channels

We use only graph presentations with $K_0 = \mathbb{Z}$ and $K_1 = \mathbb{Z}$. Start with

$$B_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad A_1 = I + B_1.$$

The cokernel identification sends $[e_1]$ to 1 and $[e_2]$ to -1; the kernel has generator $(1, -1)$. At later steps take a positive presentation $B_{h+1} = U_h B_1$ provided by Lemma 3, with the induced cokernel identification and the same kernel convention. The source is the previously chosen B_h; there is no requirement that U_h relate consecutive presentations directly.

Write $D_h, F_{j,h}, P_h$ for the corresponding finite graph triple and product. At step h apply the simultaneous lifting lemma to the three prescribed graded maps

(4)

channel	$K_0(P_h) \to K_0(P_{h+1})$	$K_1(P_h) \to K_1(P_{h+1})$
$+$	$+1$	0
$-$	-1	0
0	0	0.

Denote the resulting positive common and factor multiplicities by T_c, S_c, R_c, for $c \in \{+, -, 0\}$. With $q = q_h$, choose target sizes by

$$\begin{pmatrix} k_{h+1} \\ \ell_{h+1} \end{pmatrix} = (T_+ + T_- + qT_0) \begin{pmatrix} k_h \\ \ell_h \end{pmatrix}. \tag{5}$$

For the zero channel, use the source triple $M_q(D_h), M_q(F_{0,h}), M_q(F_{1,h})$, whose graph weights are $qk_h, q\ell_h$. Its multiplicities are T_0, S_0, R_0; hence its contribution to the target dimensions is exactly qT_0. The positive and negative channels use the unamplified source.

Place the three common representations in disjoint coordinate blocks inside D_{h+1}. Their units are orthogonal constant projections p_+, p_-, p_0 adding to one. For each factor, representation equivalence of the common algebra gives a permutation unitary matching the entire channel restriction, including the $M_q(D_h)$ restriction in channel zero. Thus there are literal maps of triples with induced product maps

$$\gamma_+ : P_h \to p_+ P_{h+1} p_+, \quad \gamma_- : P_h \to p_- P_{h+1} p_-, \quad \widetilde{\gamma}_0 : M_q(P_h) \to p_0 P_{h+1} p_0.$$

Here the full product of the amplified triple is $M_q(P_h)$. This follows either by the common matrix units or by the same tensor universal property below with coefficient M_q. All three source products are simple, and each map is nonzero, hence injective. The graded maps of $\widetilde{\gamma}_0$ under the usual corner Morita identification are zero. They are not multiplied by q: the source K_0 generator is a minimal projection of an amplified factor block.

Now tensor the triples by E. For any unital coefficient E which is nuclear, the full universal property gives

$$(F_{0,h} \otimes E) *_{D_h \otimes E} (F_{1,h} \otimes E) \cong Q_h := P_h \otimes E. \tag{6}$$

Indeed the common copy $1 \otimes E$ commutes with both graph-factor copies $F_{j,h} \otimes 1$, hence with their full product. First obtain the maximal tensor product and then use nuclearity of E. The reverse map is defined on factor generators; both composites fix all generators. The coefficient copy need not be internally commutative.

On the common and factor algebras, form the orthogonal sum of the following maps, with the matching graph component in place of γ:

$$\Phi_h(x) = (\gamma_+ \otimes \mathrm{id}_E)(x) + (\gamma_- \otimes \eta\varepsilon)(x) + (\widetilde{\gamma}_0 \circ (\mathrm{id}_{P_h} \otimes \sigma_h))(x) \otimes 1_E. \tag{7}$$

In the last term identify $P_h \otimes M_q$ with $M_q(P_h)$. All channel units lie in the common algebra and are orthogonal, so the sum is a unital *-homomorphism. The two constituent squares commute

literally, because each channel is a map of triples and its coefficient map is the same on all three algebras. Thus (7) is precisely the induced map of full products, not a separately chosen map with matching K-theory. The positive identity-coefficient channel is injective (minimal tensor products preserve embeddings), so every common, factor and product connecting map is injective.

6. Exact KK calculation, including the non-UCT coefficient

Each P_h satisfies UCT: the finite-dimensional inclusions have conditional expectations; the canonical two-sided cone is KK-equivalent to SP_h by [4, Theorem 4.1]; its extension has finite-dimensional quotient and suspended finite-dimensional ideal and therefore belongs to the Rosenberg–Schochet bootstrap category. UCT transfers under KK-equivalence and suspension by [5, Theorem 1.17, Proposition 7.1]. Together with $K_*(P_h) = (\mathbb{Z}, \mathbb{Z})$, this gives a KK-equivalence

$$\kappa_h : P_h \simeq_{KK} \mathbb{C} \oplus S\mathbb{C}$$

respecting our chosen generators. One can construct inverse classes using the UCT: its Ext term vanishes for free source K-groups, so the prescribed inverse K-maps lift uniquely and their composites are identities. The same observation shows that each even graph KK class is determined by its two integer K-maps. There are no even off-diagonal entries between $\mathbb{C}$ and $S\mathbb{C}$.

Exterior tensoring κ_h with 1_E identifies Q_h in KK with

$$X = E \oplus SE.$$

For the positive channel, the class is $\mathrm{diag}(1_E, 0)$. For the negative channel, it is $\mathrm{diag}(-[\eta\varepsilon], 0)$. The graph class of $\widetilde{\gamma}_0$ is zero in KK, since its source is Morita equivalent to the UCT algebra P_h with free K-groups and its two K-maps vanish. Its composition with any $\mathrm{id} \otimes \sigma_h$ and scalar inclusion is consequently KK-zero, even though it is an injective actual homomorphism. Hence the total map in these coordinates is

$$[\Phi_h] = f := \mathrm{diag}(e, 0), \qquad e = 1_E - [\varepsilon] \otimes [\eta]. \tag{8}$$

This argument uses no UCT or Künneth theorem for E or B.

Let $L = \varinjlim(Q_h, \Phi_h)$. We justify fully why $L \simeq_{KK} F$. In the KK category define $j : F \to X = E \oplus SE$ by $[\iota]$ in the first coordinate and zero in the second, and $s : X \to F$ by r in the first coordinate. Then

$$j \otimes s = 1_F, \qquad s \otimes j = f.$$

For each separable Z, the inverse system $\mathrm{KK}^i(X, Z)$ has bonding map precomposition with the fixed idempotent f. It splits as a constant identity system on $\mathrm{im}(f^*)$ and a zero-bonding system on $\ker(f^*)$. Thus

$$\varprojlim \mathrm{KK}^i(X, Z) \cong \mathrm{KK}^i(F, Z), \qquad \varprojlim{}^1 \mathrm{KK}^i(X, Z) = 0.$$

The second equality can also be seen directly from the product formula for $\lim^1$: $1-\mathrm{shift}$ is surjective on the zero system and on the constant identity system (solve recursively). No countability of these KK groups or compactness of F is being assumed.

All the Q_h are separable nuclear, so the natural Milnor exact sequence applies to $\mathrm{KK}^*(-, Z)$ [5, Theorems 1.12 and 1.14(b)]. It gives

$$\mathrm{KK}^i(L, Z) \cong \mathrm{KK}^i(F, Z). \tag{9}$$

Here is an explicit representing class for this isomorphism. Transport j to $j_h \in \mathrm{KK}(F, Q_h)$ using $\kappa_h \otimes 1_E$. Equation (8) implies $j_h \otimes [\Phi_h] = j_{h+1}$. If $\lambda_h : Q_h \to L$ is the canonical map, then $\beta = j_h \otimes [\lambda_h] \in \mathrm{KK}(F, L)$ is independent of h. The isomorphism (9) is precomposition with β. Surjectivity with $Z = F$ gives $\alpha \in \mathrm{KK}(L, F)$ satisfying $\beta \otimes \alpha = 1_F$; injectivity with $Z = L$ gives $\alpha \otimes \beta = 1_L$. Hence α is an invertible class. Moreover the restrictions of α to Q_h, expressed in the

$E \oplus SE$ coordinates, are exactly s. This follows either from Milnor injectivity or from $s \otimes j = f$ and the compatibility equations. Thus the retained K-class of an initial element is its reduced F coordinate in the first E summand. We will use this explicit statement to prescribe the unit of a common corner.

7. Simplicity, pure infiniteness, and stable finiteness

Let $D_\infty, A_{0,\infty}, A_{1,\infty}$ be the constituent limits. They are separable nuclear and unital, and $D_\infty \to A_{j,\infty}$ is injective. The product limit L is separable nuclear unital. All connecting maps are injective, so we regard stages as subalgebras.

We first record two details about the evaluation mechanism. A channel path in any iterated map has a support projection in the constant finite-dimensional common algebra of the target stage. These supports are orthogonal; compression by the selected support isolates that path. This follows by induction from the orthogonal common supports at each step: coefficient maps send constant projections to constant projections. Second, a path of positive identity channels commutes with coefficient evaluation by every σ. Thus, if a finite-dimensional representation σ detects a given nonzero positive stage element, following any number of identity channels and then the zero/noise channel using that same σ still gives a nonzero element. Injectivity of the evaluated graph or finite-dimensional maps proves this assertion.

Let I be a nonzero ideal in any one of the four limits. Choose a nonzero $b \in I_+$ and a positive stage approximation a, close enough that for some $\epsilon > \|a - b\|$ the cut $c = (a - \epsilon)_+$ is nonzero. Its image in the quotient by I is zero, so $c \in I$. The separating family supplies a σ with $(\mathrm{id} \otimes \sigma)(c) \neq 0$. Select a later occurrence of that representation in the tail sequence and follow identity channels until then, and the zero/noise channel at that step. Compressing by the constant path support gives a nonzero constant positive element $d \in I$.

For a constituent, at least one source matrix block evaluates nontrivially. Every entry of T_0, S_0, R_0 is strictly positive. Consequently the image d is nonzero in every target constant matrix summand. A nonzero positive matrix generates its matrix block as an ideal; hence I contains every target constant block unit and therefore the common unit. For the product, d is a nonzero positive element of the simple constant graph algebra P_{h+1}, so it generates that algebra's unit as an ideal. This also forces $I = L$. This proves simplicity of all four limits.

For pure infiniteness of L, take any $0 \neq b \in L_+$ and choose a positive stage approximation a and a nonzero cut $c = (a - \epsilon)_+ \precsim b$. The preceding evaluation procedure isolates a nonzero constant positive d at a later stage. It is an orthogonal summand of the image of c, so $d \precsim c \precsim b$. The constant graph algebra is simple purely infinite, so $\mathrm{Her}(d)$ there contains an infinite projection v. This remains infinite in L, since embeddings preserve the proper-isometry witness and its nonzero defect projection. For completeness, from $v \precsim b$ choose x so that vx^*bxv is invertible in vLv. Then

$$w = b^{1/2}xv(vx^*bxv)^{-1/2}$$

has $w^*w = v$ and $ww^* \in \mathrm{Her}(b)$. Thus $\mathrm{Her}(b)$ contains an infinite projection, as required. This proves pure infiniteness, without claiming it for the coefficient tensor stages themselves.

Every unital RFD algebra has a tracial state, by composing a nonzero unital finite-dimensional representation with normalized matrix trace. Thus the unital constituent stages, finite sums of matrix algebras over E, have nonempty compact trace spaces. For the inverse system of these spaces, every finite compatibility requirement can be met by choosing a trace at a sufficiently late stage and restricting backwards. Compactness and the finite intersection property give compatible traces on each limit, without any surjectivity assumption on trace restriction maps. Simplicity makes the limit trace faithful. Its matrix traces are faithful, so no matrix algebra over the constituent admits a proper isometry. All three constituents are stably finite.

Finally, the literal diagrams imply the full universal identity

$$L \cong A_{0,\infty} *_{D_\infty} A_{1,\infty}. \tag{10}$$

Compatible factor maps give a map from the displayed full product to L. Conversely, the two stage factors map compatibly into the displayed full product, giving compatible maps from Q_h and therefore from L. The two composites fix every stage-factor generator and are identities. All maps preserve the common unit.

8. Prescribing the unit inside the common algebra

The preceding construction identifies L with B up to KK-equivalence, but an arbitrary unit must still be obtained by an actual common corner. Let $u \in K_0(F)$ be the inverse image of $[1_B]$ under ξ_*. By the definition of nonunital K_0, its image in $K_0(E)$ can be written

$$\iota_*(u) = [p_0] - [q_0], \qquad p_0, q_0 \in M_t(E)$$

with equal scalar ranks under ε. Stabilize so that the matrices have the same size; add the same nonzero scalar projection to both if necessary to ensure the resulting common projection below is nonzero. Choose the initial weights $k_{1,1}, k_{1,2}$ large enough to contain these two matrix projections. Choose positive $\ell_{1,1}, \ell_{1,2}$ so that

$$k_{1,1} + \ell_{1,1} = k_{1,2} + \ell_{1,2}.$$

Then $[1_{P_1}] = 0$ in $\operatorname{coker} B_1 = \mathbb{Z}$, though the algebra is unital. All later weights are chosen by (5) as before.

Put the actual projection p in $D_1 \otimes E$ by placing p_0 in the first k-block, q_0 in the second k-block, and zero elsewhere. The inclusion into $P_1 \otimes E$ sends these two blocks to the positive and negative $K_0(P_1)$ generators. Thus, in the first E coordinate of the KK identification $Q_1 \simeq E \oplus SE$, its class is $[p_0] - [q_0] = \iota_*(u)$, and its second coordinate is zero. This follows also by external products with the two minimal graph projections; their even KK classes are $+1$ and -1 in the scalar component and zero in the suspension component. The calculation does not assume a Künneth formula for E.

The KK equivalence $\alpha : L \to F$ constructed above consequently sends $[p]$ to u. Since all maps are injective, p is a nonzero projection of the simple D_∞, hence full there. It is also full in the other three simple limit algebras.

Lemma 5 (Common full corner)**.** *If $P = A_0 *_D A_1$ and $p \in D$ is a full projection, then*

$$pPp \cong (pA_0p) *_{pDp} (pA_1p)$$

as a full unital amalgam with unit p.

Proof. Fullness supplies finitely many $x_r \in Dp$ with $\sum_r x_r x_r^* = 1$: normalize a finite invertible positive sum from the ideal generated by p. Set $R = (x_r^* x_s) \in M_N(pDp)$, a projection. The maps $a \mapsto (x_r^* a x_s)$ identify A_j with $RM_N(pA_jp)R$ and agree on D. If $T = (pA_0p) *_{pDp} (pA_1p)$, the full property gives $\Psi : P \to RM_N(T)R$. The column $z = (x_r^* p)_r$ satisfies $z^*z = p$ and $zz^* = \Psi(p)$. Thus $a \mapsto z^*\Psi(a)z$ maps pPp to T and fixes its two factor corners. It is inverse to the natural map $T \to pPp$. To verify generation of pPp by these corners, insert $1 = \sum_r x_r x_r^*$ between successive letters of each compressed alternating word; every resulting compressed factor lies in the appropriate pA_jp. The dense word algebra proves both composites are identities. □

Apply the lemma to (10). The three corners are separable nuclear simple unital stably finite, and pLp is a unital Kirchberg algebra. Its full-corner inclusion into L is a KK-equivalence. Compose that class with α and ξ to obtain an invertible element of $\mathrm{KK}(pLp, B)$ taking its unit $[p]$ to $[1_B]$. Phillips [6, Corollary 4.2.2] gives a unital isomorphism $pLp \cong B$. This precise corollary requires no UCT hypothesis. The common-full-corner identity now proves Theorem 1.

9. Scope and formalization

If correct, Theorem 1 answers Suzuki's question for every unital Kirchberg algebra, with nuclear constituents. It imposes no UCT hypothesis and no restriction on the target's K-groups. The coefficient algebra retains the target's full KK information; the argument does not establish the UCT for arbitrary nuclear C*-algebras. It also does not assert that the three constituents can always be chosen AF or AT.

The accompanying theorem `Suzuki.fullTheoremConditional` derives the exact main claim from explicit `PublishedInputs`. This is conditional Lean verification: the construction, KK calculation, limits and prescribed unit are derived within those interfaces. For stable finiteness, Lean uses faithful matrix evaluations and an injective matrix limit; the written argument uses compatible traces.

External inputs include standard graph C*-algebra structure and K-theory; KK products, split exactness, mapping cones, Bott and Morita equivalence; UCT for scalar bootstrap graph stages; the analytic Milnor sequence; Dadarlat's published extension construction; standard nuclearity, lifting, tensor and Cuntz-comparison facts; and unit-preserving Kirchberg classification with its exact hypotheses. No UCT is assumed for the target or coefficient algebra.

Realizing the interfaces in actual Kasparov KK theory and matching the published results remain external obligations; kernel audits do not discharge them. Exact statements, source versions, proof uses and access limitations are recorded in the verification notes and dependency ledger. No unconditional formalization is claimed. The mathematical argument is unchanged from the first version.

Source note. The cited statements of Rosenberg–Schochet were consulted through text extracted from the original journal scan; the complete original PDF was not available for direct inspection. The bibliography identifies the source versions used, and theorem and page numbers refer to those versions.

References

[1] Y. Suzuki, *Free groups amenably act on unital simple AF-algebras*, arXiv:2610.01460v1, 1 October 2026. Corollary 1.4 and Remark 1.5.

[2] A. Kumjian, D. Pask and I. Raeburn, *Cuntz–Krieger algebras of directed graphs*, February 1998 author version, author PDF. Theorem 1.2 and Corollary 3.11 (PDF p. 14). Published in Pacific J. Math. **184** (1998), 161–174.

[3] P. Ara and K. R. Goodearl, *C*-algebras of separated graphs*, arXiv:1102.4296v2, 11 July 2011. Remark 3.10 records nuclearity of ordinary row-finite graph algebras; separated-graph nuclearity is not asserted.

[4] P. Fima and E. Germain, *The KK-theory of amalgamated free products*, arXiv:1510.02418v3, 27 December 2016. Introduction p. 2; Theorem 4.1 p. 20; cone extension p. 25.

[5] J. Rosenberg and C. Schochet, *The Künneth theorem and the universal coefficient theorem for Kasparov's generalized K-functor*, Duke Math. J. **55** (1987), 431–474. DOI. Theorems 1.12, 1.14(b), 1.17; Proposition 7.1. See the source note above.

[6] N. C. Phillips, *A classification theorem for nuclear purely infinite simple C*-algebras*, arXiv:funct-an/9506010v2, 18 December 1997. Corollary 4.2.2, p. 42 (no UCT hypothesis).

[7] M. Dadarlat, *Some remarks on the universal coefficient theorem in KK-theory*, January 2002 author version, author publication list. Lemma 2.4, pp. 2–3; construction in the proof of Theorem 1.2, p. 8. Published in *Operator Algebras and Mathematical Physics*, Theta (2003), pp. 65–74.